Closing the paired interior between two singleton ends

Human Compression project: L044 continuation

Use all hypotheses and notation of the opposite-singleton budget packet: $G \le A_n$, $n = 2r \ge 14$, has a connected transitive string C-marking, all maximal omissions are intransitive, and

$$G_0 = \mathrm{Alt}(\Omega \setminus \{a\}), \qquad G_{r-1} = \mathrm{Alt}(\Omega \setminus \{d\}).$$

Every colour $2, \dots, r-2$ has two ambient-eligible edges. We additionally assume the alternating rank bound at strictly smaller degrees at least twelve. The complete accepted primitive, anchored imprimitive and small-tail bounds remain paid. Write $\sigma = \rho_1$, $\eta = \rho_{r-1}$, $H_C = G_{\{2,\dots,r-2\}}$, $K_C = G_{\{3,\dots,r-2\}}$, and $K = G_{>2}$, as in the predecessor.

Lemma 1 (The actual two short heads)**.** *After renaming points, one of the following two literal pairs occurs, with both generators fixing every point not displayed:*

	ρ_0	ρ_1
five-point head	$(a\ b)(u\ v)$	$(b\ u)(v\ w)$
six-point head	$(a\ b)(u\ v)$	$(b\ x)(u\ w)$.

(1)

The first pair generates the natural five-point D_{10}; the second generates S_3 diagonally on two three-point orbits. The group K fixes the displayed five or six points, respectively. Moreover colour 1 has at least two ambient-eligible edges.

Proof. We use the proved cases of the opposite-singleton budget, not an unproved stronger interpretation of its statement. That proof excludes $f_C = 3$ and allows only $f_C = 4$ or its literal residual $f_C = 2$. Here f_C counts the H_C-fixed points in the G_{r-1}-domain.

If $f_C = 4$, then H_C fixes a, b, u, v and is transitive on $V = \Omega \setminus \{a, b, u, v, d\}$. Since $C_0 = \langle \sigma, H_C \rangle$ is transitive on its complement of a, d, the involution σ must send each of b, u, v into V. A fixed singleton or a transposition among these three would instead be a proper C_0-orbit. Their three images are distinct. The subgroup K_C fixes a, b, u, v and their images under σ, since it centralizes σ. Together with d this gives eight ambient fixed points. Its $r-4$ colours all have two eligible edges, confined to at most $n-8 = 2(r-4)$ points. The strict localized interval bound contradicts this, retaining fixed singleton orbits as usual. Thus $f_C = 4$ is impossible.

Consequently the predecessor's $f_C = 2$ residual holds: $E = \{u, v\}$ is a transitive H_C-orbit, V is its other nontrivial orbit of odd size $n-5$, and $\rho_0 = (a\ b)(u\ v)$. Its proof further establishes that K_C fixes E pointwise, that η fixes E pointwise, and that $x = b^\sigma \ne b$ lies in $E \cup V$. If $x \in E$, name it u. Then $u^\sigma = b$ and $v^\sigma = w \in V$, exactly as proved there. This gives the first pair in (1), initially with a possible extra involution ϵ in σ outside the displayed five points.

If $x \in V$, then $E^\sigma \ne E$ by transitivity of C_0. It cannot be disjoint from E: in that case K_C fixes the seven local points a, b, x, E, E^σ, and the same strict bound on its $r-4$ colours is impossible. Their intersection is therefore a singleton. Since σ is an involution, that singleton is fixed by σ; name it v. Thus $u^\sigma = w \in V \setminus \{x\}$. This is the second pair in (1), again initially decorated by an exterior involution ϵ.

In either case the two displayed transpositions of σ are even, so ϵ is even. In the first case $(\rho_0\sigma)^5 = \epsilon$, and in the second $(\rho_0\sigma)^3 = \epsilon$. The displayed product is respectively a five-cycle or two disjoint three-cycles. Hence

$$\epsilon \in \langle \rho_0, \rho_1 \rangle \cap G_0 = \langle \rho_1 \rangle,$$

where membership in G_0 follows from its full natural alternating action and ϵ fixing a. The decoration fixes b whereas ρ_1 moves it, so $\epsilon = 1$. This proves the exact marked heads without quotient IP or a finite classification. The predecessor's fixed-point argument shows that K fixes all five or six displayed points.

Finally H_C preserves E and V, while η fixes E, a, b and moves d into V. Thus the actual omission $G_1 = \langle \rho_0, H_C, \eta \rangle$ preserves the three nonempty disjoint sets $\{a, b\}$, E, and $V \cup \{d\}$. It has at least three orbits. A single eligible 1-edge could join at most two of them when ρ_1 is adjoined, contradicting transitivity of G. Colour 1 is therefore paired too. □

Lemma 2 (Both trimmed tails have an orbit deficit)**.** *The group $K = G_{>2}$ is intransitive on its nonempty support and fixes at least five points. The same holds for $G_{<r-3}$.*

Proof. Suppose K is transitive on its support X, of size m. It has rank $e = r - 3 \geq 4$, an actual even string C-marking and a connected diagram. Its local maximal omissions are intransitive: if one were transitive on X, adjoining the omitted generator, which is supported on X, could not change the global omission orbits, contradicting global transitivity. All generators except ρ_2 preserve X, since the earlier ones centralize K. Thus ρ_2 takes some point of X outside, and that point is fixed by all marks of K except its first. This is the exact attachment anchor required by the accepted bounds.

The complete accepted small-tail lemma leaves only the five-point D_{10} or six-point A_5 markings when $m < 12$ and $e \geq (m-1)/2$. Their ranks are at most three, excluded here. For $m \geq 12$, the accepted proper-primitive and anchored-imprimitive bounds, and the explicit smaller-degree induction when $K = \mathrm{Alt}(X)$, give

$$2e \leq m - 1.$$

The same inequality holds in the small case just discussed. Since Lemma 1 supplies at least five fixed points, $m \leq n - 5 = 2e + 1$. Equality is therefore forced throughout: $m = n - 5$, and the fixed set consists of exactly five points. The six-point head is excluded. In the five-point case the two actual G_2-orbits are its head D and X, because $G_2 = \langle \rho_0, \rho_1, K \rangle$ has these disjoint transitive factors. The generator ρ_0 fixes X and has just one fixed point, w, in D. As ρ_2 commutes with ρ_0, any 2-edge between D and X must have endpoint w. There is at most one such edge, contradicting the assumed two eligible 2-edges. Thus K is intransitive on its support.

Lemma 1 proved that every interior colour $1, \ldots, r-2$ is paired. Apply the whole argument in reversed order; all its precise hypotheses hold. It gives the assertion for $G_{<r-3}$. □

Theorem 3 (Two singleton ends with a paired interior)**.** *There is no group satisfying the stated hypotheses.*

Proof. Apply the accepted orbit-incidence deficit theorem with $h = 2$ and $g = r - 3$. Since $r \geq 7$, we have $g > h$. The two groups $R = G_{>h}$ and $P = G_{<g}$ are intransitive on their supports by Lemma 2, and each fixes at least five points. Thus both required outside budgets hold:

$$2h = 4 \leq n - |\operatorname{supp}(R)| - 1, \qquad 2(r - 1 - g) = 4 \leq n - |\operatorname{supp}(P)| - 1.$$

Every middle colour $h < j < g$ is paired. With $p, q \geq 2$ the numbers of their nontrivial orbits, the complete incidence theorem gives $2r \leq n + 3 - p - q \leq n - 1$, contradicting $n = 2r$. □

Attribution and exact remaining source boundary. This closes the stated subcase of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.32, physical pages 44–45. The source's two-end fixed-support and overlap strategy is retained. All accepted odd singleton, opposite-end lift, selected-localization, anchored-tail and orbit-incidence proofs remain paid. Short-head extraction uses actual IP and the natural alternating omission. Other heads, unpaired intermediate crossings, equality diagrams and remaining rigidity/table obligations are separate. No new census, formalization, original or question credit follows.